autStand
Automation Standard

Commissioning Tool

Operator & Administrator Guide

CENTRAL CONTROL · FIELD TESTING · PLC PROGRAMMING

Field commissioning system · Central hub + offline field client

CONTENTS

What's inside

SECTION 1

Overview & how it fits together

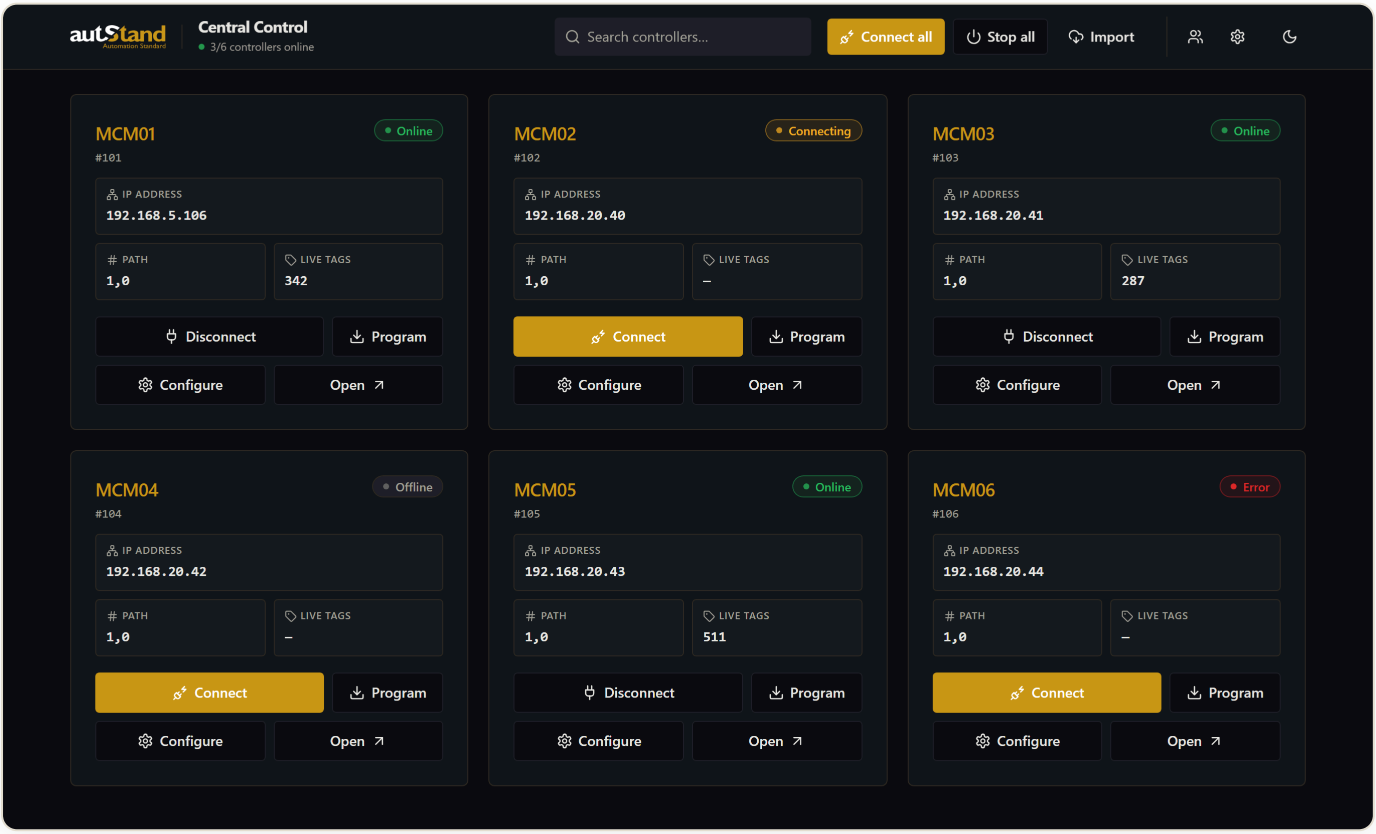
The Commissioning Tool is a field system for bringing PLC-controlled machines online. It connects to Allen-Bradley controllers over Ethernet/IP, streams live tag states, lets technicians mark every I/O point pass/fail, and syncs results to the central cloud.

Two faces, one data model

- **Central Control hub** — a single screen listing every controller (MCM) in the project. Connect, configure, and program any of them from one place.
- **Field commissioning client** — the offline-first testing surface for a single subsystem: the live I/O grid, pass/fail marking, and Guided Mode.

Both run from the same application. A technician opens the tool's address in a browser, picks a controller from the hub, and works. Everything is stored locally first (so a dropped network never loses data) and synced to the cloud in the background.

> **Who uses it.** Field technicians connect and test controllers; administrators set the cloud project key, manage accounts, and program controllers. Roles are explained in Section 8.

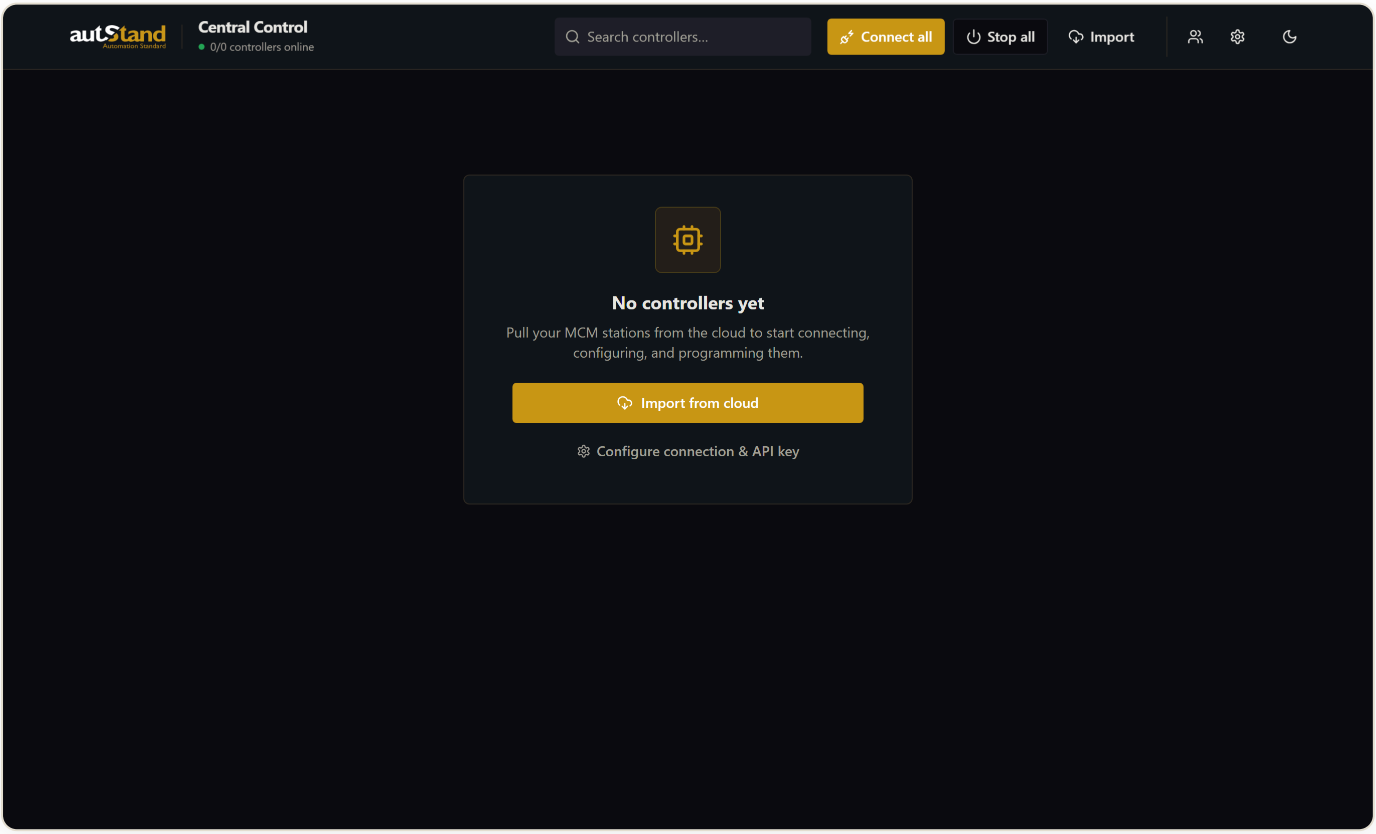


The Central Control hub — every controller in the project, with live status.

Getting started

The tool runs on one machine (the “server laptop”). Everyone else just opens its address in a browser — the same address, every time.

1. **Open the tool.** In a browser go to `http://<server-ip>:3000/mcm`. You land on the Central Control hub.
2. **Connect to the cloud project.** First run only: an admin pastes the project API key (Section 8). This tells the tool which project it belongs to.
3. **Import the controllers.** Click **Import** — every controller (MCM) in the project appears as a card.
4. **Work.** Pick a controller, Connect, and start commissioning.



Before any controllers are imported, the hub guides you to Import from cloud.

> **No controllers showing?** The project API key isn't set yet, or Import hasn't been run. See Section 8.

SECTION 3

The Central Control hub

The hub is the home screen: one card per controller, each showing its live status, address, and the actions you can take.

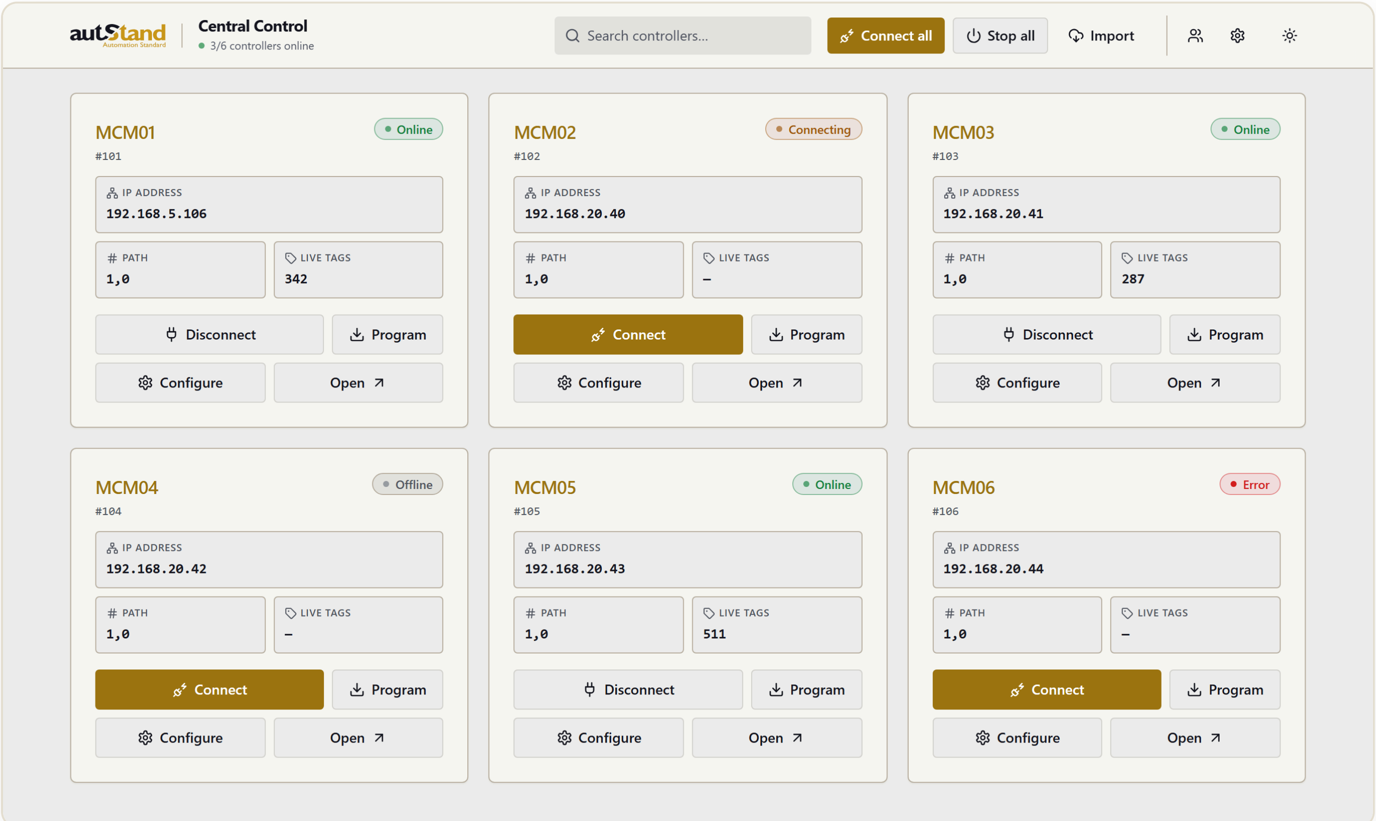
Reading a card

Every card shows the controller name, its subsystem id, IP address, backplane path, and live tag count. The status badge tells you its state at a glance:

Online connected & streaming tags **Connecting** reconnecting **Error** connection problem **Offline** not connected

The header

- **Search** — filter the grid by name, id, or IP instantly.
- **Connect all / Stop all** — bring the whole fleet up or down at once.
- **Import** — pull the controller list from the cloud project.
- **Gear / People icons** — Settings (cloud key, controllers) and Accounts.



The hub adapts to light and dark themes; the brand gold stays consistent.

SECTION 4

Connecting to a controller

Once a controller's address is set, connecting is one click — no typing, for anyone.

1. Find the controller's card on the hub.
2. Click **Connect**. The status badge turns **Online** and the live tag count starts climbing.
3. To take it offline, click **Disconnect**. Use **Stop all** in the header to disconnect the whole fleet.

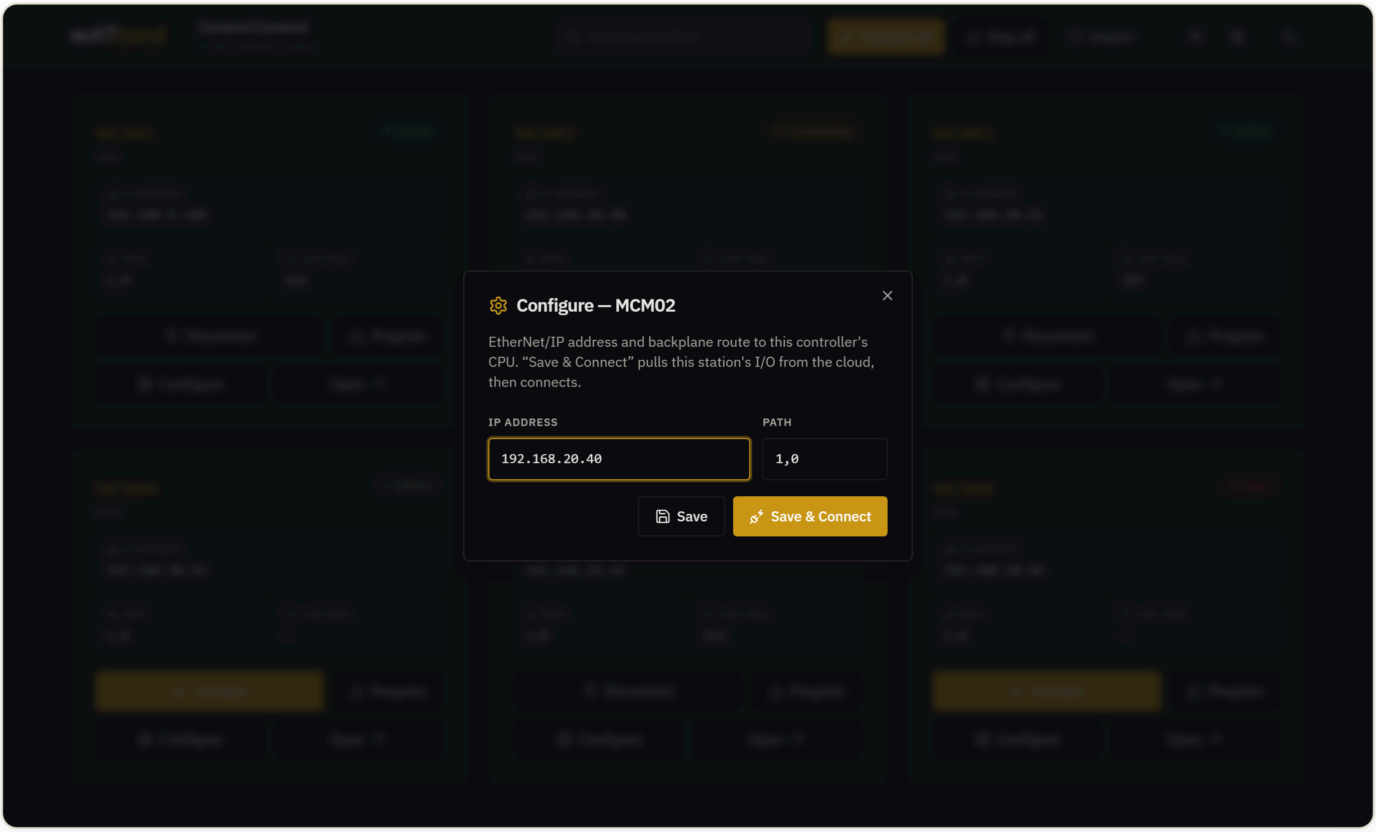
> **Why no IP typing?** Each controller's IP and path are stored on the server once (Section 5). After that, every technician who opens the hub just clicks Connect.

SECTION 5

Configuring a controller

Setting a controller's connection is a one-time job. After it's saved, it's shared by everyone.

1. On the controller's card, click **Configure**.
2. Enter the **IP address** and the **Path** (the backplane route to the CPU — commonly `1,0`).
3. Click **Save** to store it, or **Save & Connect** to store it, pull the station's I/O from the cloud, and connect in one step.



The Configure dialog: just an IP and a path. Saved once, shared by all.

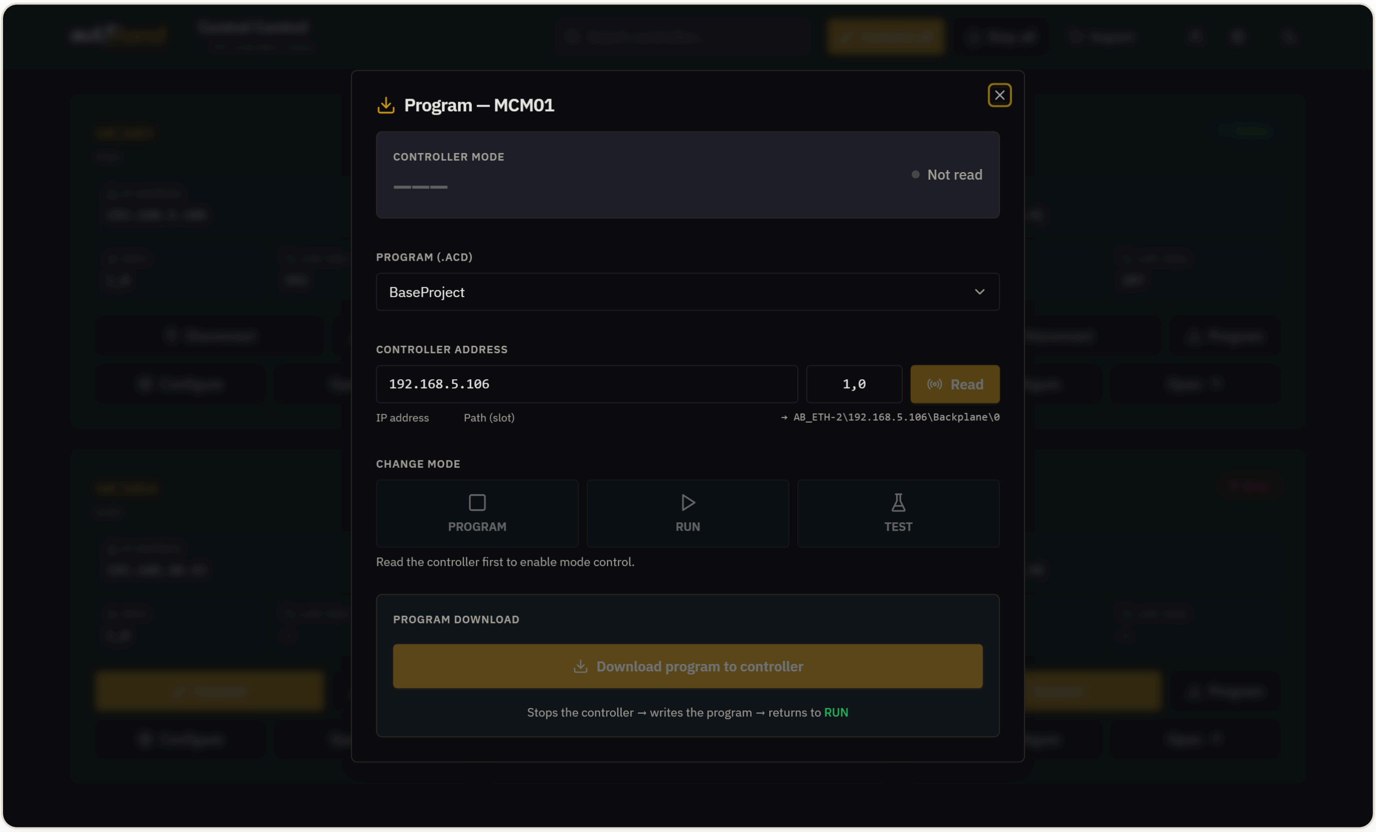
> **Path tip.** The path is the route across the chassis backplane to the controller CPU — usually `1,0` (port 1 = backplane, slot 0 = CPU).

SECTION 6

Programming a controller

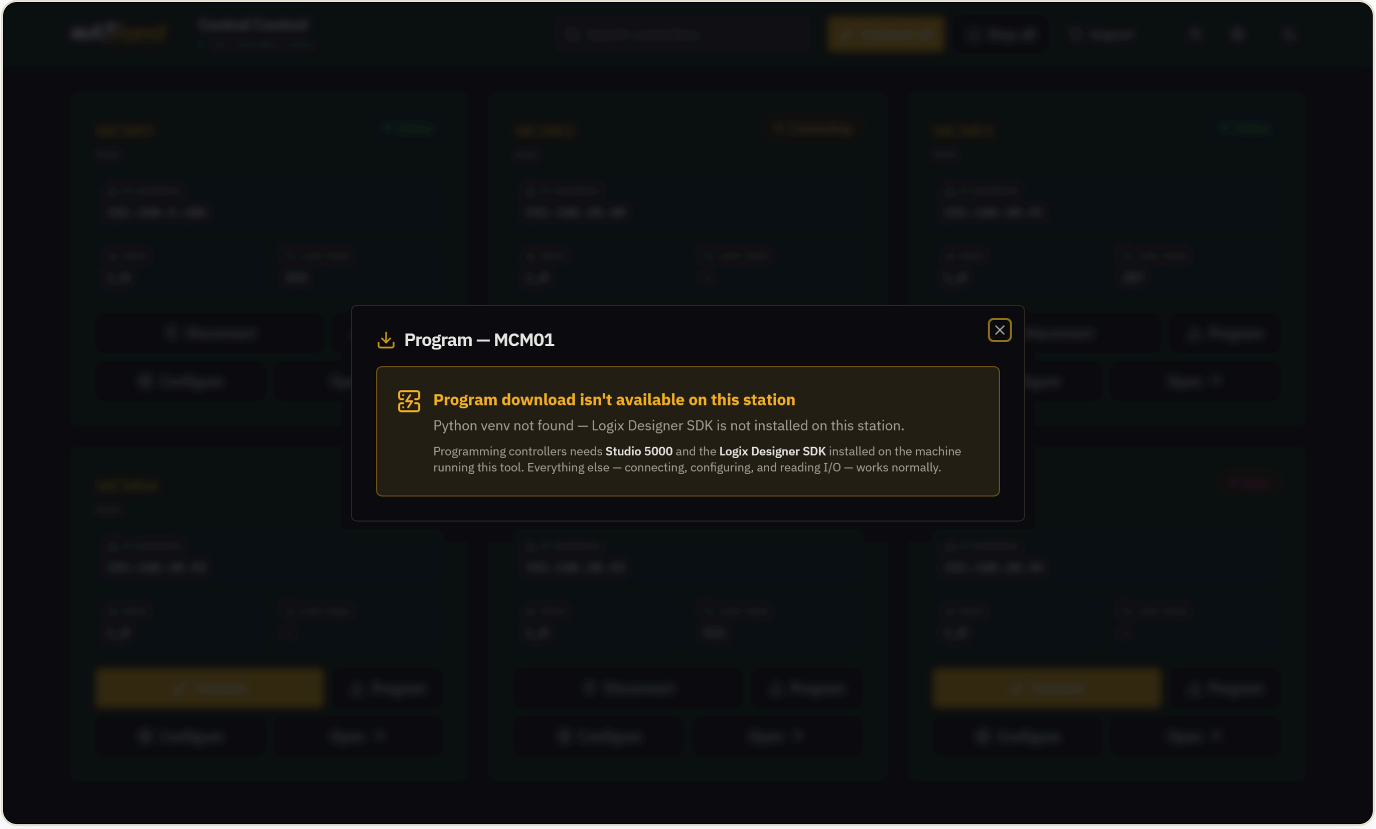
Download a program to any controller directly from the hub — no separate engineering workstation hunt.

1. On the controller's card, click **Program**.
2. Pick the **project (.ACD)** from the list of programs on this station.
3. Confirm the **IP address** and **Path** (pre-filled from the controller's config). The resolved communications path is shown beneath for transparency.
4. Optionally **Read** the controller to see its current mode, or switch mode (Program / Run / Test).
5. Click **Download program to controller**. The tool stops the controller, writes the program, and returns it to **RUN** — with a live progress pipeline.



The Program dialog: pick a project, confirm IP + path, download. Mode control and a staged progress pipeline are built in.

Requires Studio 5000 + the Logix Designer SDK on the machine running the tool. On a station without them, the Program dialog cleanly reports it's unavailable — and every other feature (connect, configure, test) keeps working normally.



Graceful degradation: on a station without the SDK, programming is clearly marked unavailable — nothing breaks.

SECTION 7

Commissioning & Guided Mode

From a controller's card, **Open** takes you into the field commissioning client for that subsystem.

The I/O grid

- Every I/O point for the subsystem is listed with its **live state** streamed from the PLC.
- Technicians mark each point Pass or Fail; failures capture a reason and responsible party.
- Results save locally first, then sync to the cloud — so a dropped link never loses work.

Guided Mode

Guided Mode walks a technician through commissioning **one step at a time** instead of facing the whole grid at once. It groups work into phases → segments → tasks → steps, auto-detects I/O checks against the live PLC, and lets a technician skip a step with a recorded reason when something isn't ready.

- **Priority-driven** — the next most important task surfaces automatically.
- **Auto-detect** — when the PLC confirms a check, the step advances on its own.
- **Accountable skips** — skipping requires a reason, preserved in the audit trail.

> **When to use Guided Mode.** Use it for first-time commissioning and for less-experienced technicians; use the full grid for fast re-checks and spot fixes.

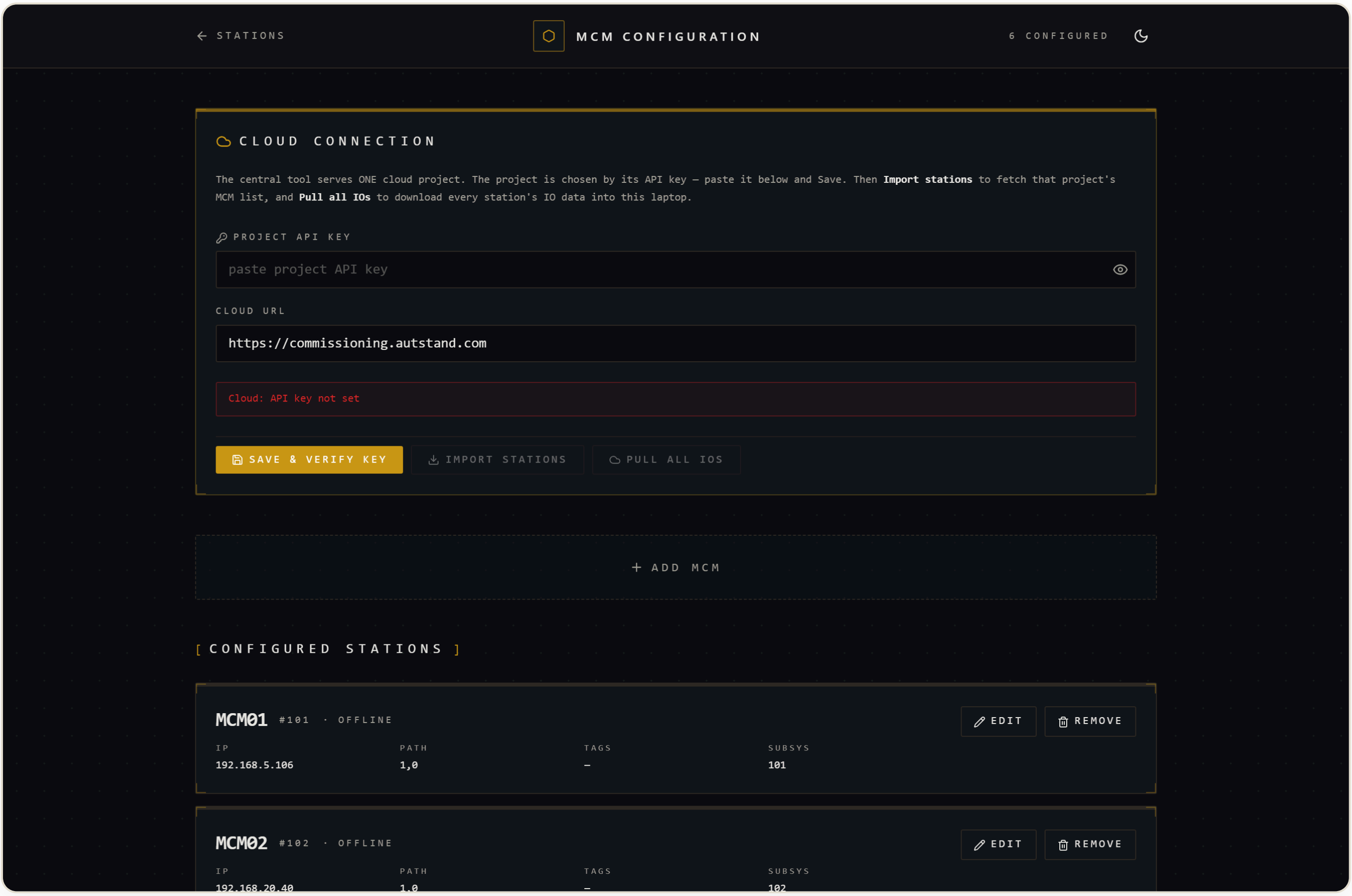
SECTION 8

Settings & administration

Choosing the project (cloud API key)

The tool serves **one cloud project**, chosen by its API key. This is also where the project is “selected”.

1. Click the **gear icon** in the hub header → opens `/settings/mcms`.
2. In **Cloud Connection**, paste the **Project API Key** and **Save**. The panel confirms “Active project: <name> — N stations”.
3. Click **Import stations** to pull the controller list, then **Pull all IOs** to download each station's I/O.



Settings — Cloud Connection. The API key both authenticates and selects the project.

Accounts & roles

- **Administrators** set the cloud key, manage accounts, configure controllers, and program PLCs.
- **Technicians** connect controllers and record test results.
- Accounts are managed from the **People icon** in the hub header (create, enable/disable, reset PIN).

> **One key = one project.** To work on a different project, switch the API key. (A single tool instance is intended to serve one project.)

SECTION 9

Sync & offline operation

The tool is offline-first by design — the field is no place to lose data to a flaky network.

- **Local first.** Every result, comment, and reset is written to the local database immediately.
- **Instant push.** Right after a local save, the tool pushes it to the cloud (typically 1–2 seconds).
- **Background retry.** Anything that fails to sync is retried automatically until it lands.
- **Authority.** The local database is the source of truth for test results; the cloud is the receiver. Both keep a full audit trail.

> **What this means for you.** Keep testing even with no network. When connectivity returns, your work syncs on its own — nothing to re-enter.

SECTION 10

Troubleshooting

Symptom	Likely cause & fix
Hub is empty / “No controllers”	Project API key not set, or Import not run. Settings → Cloud Connection → paste key → Import (Section 8).
Connect fails	Wrong IP or path on the card. Open **Configure** and correct them. Confirm the controller is powered and on the network.
Connected but no I/O	The station's I/O hasn't been pulled. Use **Save & Connect**, or Settings → **Pull all IOs**.
“Program download isn't available”	This station doesn't have Studio 5000 + the Logix Designer SDK. Program from a station that does; all other features still work.
Cloud rejected the API key	The key is wrong or for a different environment. Re-paste the correct project key in Settings.
Status stuck on “Connecting”	Network path to the PLC is intermittent. Check cabling/VPN; the tool auto-reconnects when the path is stable.

SECTION 11

Quick reference

Everyday actions

Goal	Where
See all controllers	Hub — `/mcm`
Connect / disconnect one	Card → **Connect** / **Disconnect**
Set a controller's IP/path	Card → **Configure**
Download a program	Card → **Program**
Test a subsystem's I/O	Card → **Open**
Set the project / API key	Header gear → **Cloud Connection**
Manage accounts	Header people icon

Status colors

Online connected **Connecting** reconnecting **Error** problem **Offline** not connected

> **One address to remember.** `http://<server-ip>:3000/mcm` — the Central Control hub. Everything starts there.